DESIGN COMPUTED · FREE GUIDE

Don't build the roof, hang it: the Colosseum's retractable sail roof

The free guide to the reel on the Colosseum's velarium: how masts, ropes and canvas covered a stadium, why a hanging roof beats a built one, how the crowd got out fast, and a string-and-cloth model to make.



A stadium with a roof you could pull

The Colosseum was built in the years around AD 70 to 80 and held roughly 50,000 spectators (an ancient calendar of 354 gives 87,000, but modern estimates are around 50,000). To protect them from sun and rain it had a retractable awning, the velarium.

The velarium was a canvas-covered, net-like structure of ropes with a hole in the centre, held by 240 mast corbels around the top of the attic. It sloped towards the centre and covered about two thirds of the arena.

Why hang a roof instead of building one

A fixed roof over an oval of this width would need enormous beams or a dome. A net hung from the rim only needs the rim to carry pull, and the canvas can be rolled up when the weather turns or the show is over.

In the reel's diagram: masts rise from the top of the wall, ropes run from them to a ring above the centre, and canvas strips slide along the ropes. The hole in the middle lets hot air out.

Getting everyone out

The Colosseum had eighty ground-level entrances, 76 of which ordinary spectators used. Vaulted passages (vomitoria) let the building empty within a few minutes.

Built examples

Colosseum, Rome (c. AD 70-80)	Velarium with 240 mast corbels, covering about two thirds of the arena.
Scale models of the velarium	Modern models show the rope net and red canvas strips radiating from the central opening.
Modern tensile roofs	The same idea survives in stretched-fabric stadium roofs, where the cables do the carrying.

Try it yourself: a velarium from string and cloth

1. Cut the bottom out of a round cake tin or bowl: this is your arena. Stick 12 toothpicks upright around the rim.
2. Tie a small ring (a washer or a paper ring) loosely above the centre with thread running to each toothpick top.
3. Cut 12 strips of tissue paper or cloth and tape each along one thread, from rim to ring.
4. Pull the strips toward the ring one at a time and watch the arena fill with shade.
5. Blow gently from the side: notice how the hole in the centre keeps the net from being lifted.

The rule to remember

If the span is huge, hang the surface from the edge and let ropes carry it in tension.

Default answer	Build a roof that stands on its own.
Form-finder answer	Hang the roof from the rim and make it retractable.
Check	Ask: which part of my roof could be a net instead of a slab?

Go further

Colosseum (Wikipedia)
https://en.wikipedia.org/wiki/Colosseum

Velarium (Wikipedia)
https://en.wikipedia.org/wiki/Velarium

Parco archeologico del Colosseo
https://colosseo.it

Credits
Cover photo (scale model of the velarium): Tecnoco73, CC BY-SA 4.0, Wikimedia Commons.
Facts: Wikipedia (Colosseum).

Made by Design Computed · Follow for more · @designcomputed